## 人工智能 A 2023-2024 学年期末考试真题

1. (10 points) Given numeric truth values.

$$\mu(A) = 0.4/0.5 + 0.7/0.8 + 0.6/0.9$$
$$\mu(B) = 0.5/0.5 + 0.6/0.8 + 0.7/0.9$$

calculate the fuzzy logic truth of the following:

(1) NOT A

(2) A AND B

(3) A OR B

(4) A → B

(5) B → A

**Answer:**

(1) 0.6/0.5 + 0.3/0.8 + 0.4/0.9

(2) 0.4/0.5 + 0.6/0.8 + 0.6/0.9

(3) 0.5/0.5 + 0.7/0.8 + 0.7/0.9

(4) 0.6/0.5 + 0.6/0.8 + 0.7/0.9

(5) 0.5/0.5 + 0.7/0.8 + 0.6/0.9

2. (15 points) Given information about the name, age, nationality of a person from a group using the following deftemplate,

(deftemplate person (slot name)
(slot age)
(slot occupation)
(slot nationality)

For example, we can write a rule to identify the people who is from France:

(defrule example (person (name ?nm) (nationality France) ) => (printout t ?nm) )

(a) (5 points) Write a rule that matches persons who are from U.S. and are computer programmers.
(b) (5 points) Write a rule that matches persons whose age is between 20 and 25 (inclusive) and whose nationality is either ″French″ or ″German″ .
(c) (5 points) Write a rule that modifies the age of persons who are ″Chinese″ to increase it by 5 years.

**Answers:**

(1) (defrule A (person (name ?nm) (occupation computer programmer) (nationality U.S.)) => (printout t ?nm " is from the U.S. and is a computer programmer." ) )

(2) (defrule B (person (name ?nm) (age ?a&:(>= ?a 20) &:(<= ?a 25)) (nationality ?nat&:(or (eq ?nat French) (eq ?nat German)))) => (printout t ?nm " is aged between 20 and 25 and is from " ?nat "." crlf) )

(3) (defrule increase-age-for-Chinese ?person <- (person (nationality Chinese) (age ?a)) => (modify ?person (age (+ ?a 5))) )

3. (10 points) Use resolution to solve the following problem.

1) Every investor bought stocks or bonds.

2) If the Dow-Jones Average crashes, then all stocks that are not gold stocks fall.

3) If the T-Bill interest rate rises, then all bonds fall.

4) Every investor who bought something that falls is not happy.

Prove this conclusion: If the Dow-Jones Average crashes and the T-Bill interest rate rises, then any investor who is happy bought some gold stock.
(Hints: For example, "Every child loves candy" can be written as $\forall x(\text{Child}(x) \rightarrow \text{Love}(x, \text{candy}))$, i.e., $\neg\text{child}(x) \vee \text{Love}(x, \text{candy})$. "Dow-Jones Average crashes" and "T-Bill interest rate rises" can be represented by "DJCrash" and "TBRise" .)

**Answer:**

1) $\forall x(\text{Investor}(x) \rightarrow \exists y((\text{Stock}(y) \vee \text{Bond}(y)) \wedge \text{Buy}(x, y)))$

2) $\text{DJCrash} \rightarrow \forall x((\text{Stock}(x) \wedge \neg\text{Gold}(x)) \rightarrow \text{Fall}(x))$

3) $\text{TBRise} \rightarrow \forall x(\text{Bond}(x) \rightarrow \text{Fall}(x))$

4) $\forall x \forall y(\text{Investor}(x) \wedge \text{Buy}(x, y) \wedge \text{Fall}(y) \rightarrow \neg\text{Happy}(x))$
or $\forall x((\exists y(\text{Investor}(x) \wedge \text{Buy}(x, y) \wedge \text{Fall}(y))) \rightarrow \neg\text{Happy}(x))$

Neg conclusion:

$$\neg((\text{DJCrash} \wedge \text{TBRise}) \rightarrow \forall x(\text{Investor}(x) \wedge \text{Happy}(x) \rightarrow \exists y(\text{Gold}(y) \wedge \text{Buy}(x, y))))$$

Converted to clause form:
1) $(\neg\text{Investor}(x) \vee \text{Stock}(f(x)) \vee \text{Bond}(f(x))) \wedge (\neg\text{Investor}(x) \vee \text{Buy}(x, f(x)))$
2) $\neg\text{DJCrash} \vee \neg\text{Stock}(x) \vee \text{Gold}(x) \vee \text{Fall}(x)$
3) $\neg\text{TBRise} \vee \neg\text{Bond}(z) \vee \text{Fall}(z)$
4) $\neg\text{Investor}(x) \vee \neg\text{Buy}(x, y) \vee \neg\text{Fall}(y) \vee \neg\text{Happy}(x)$

Neg conclusion (as clauses):

$$\text{DJCrash} \wedge \text{TBRise} \wedge \text{Investor}(a) \wedge \text{Happy}(a) \wedge (\neg\text{Gold}(y) \vee \neg\text{Buy}(a, y))$$

The resolution procedure is omitted.

4. (10 points) Considering the following database of dogs represented by 7 training examples. The target is to classify dangerous or well-behaved dogs, based on the other 3 attributes (Color, Fur, Size) of the dog. Construct the decision tree from the following examples and show the value of the information gain for each candidate attribute at each step in the construction of the tree.

$$\log_2\left(\frac{4}{7}\right) = -0.807, \quad \log_2\left(\frac{3}{7}\right) = -1.222, \quad \log_2\left(\frac{1}{3}\right) = -1.585, \quad \log_2\left(\frac{2}{3}\right) = -0.585, \quad \log_2\left(\frac{3}{4}\right) = -0.415.$$

| Dog | Color | Fur | Size | Class |
|---|---|---|---|---|
| 1 | brown | ragged | small | well-behaved |
| 2 | black | ragged | big | dangerous |
| 3 | black | smooth | big | dangerous |
| 4 | black | curly | small | well-behaved |
| 5 | white | curly | small | well-behaved |
| 6 | white | smooth | small | dangerous |
| 7 | brown | ragged | big | well-behaved |

Answer:

**Fur** - ragged - Color - brown: well-behaved - black: dangerous - smooth: dangerous - curly: well-behaved

**Step 1:**

$$H(D) = -\left(\frac{3}{7} \times \log_2\left(\frac{3}{7}\right) + \frac{4}{7} \times \log_2\left(\frac{4}{7}\right)\right) \approx 0.985$$

**Color:**

$$H(D_{color=brown}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D_{color=white}) = -\left(\frac{1}{2} \times \log_2\left(\frac{1}{2}\right) + \frac{1}{2} \times \log_2\left(\frac{1}{2}\right)\right) = 1$$

$$H(D_{color=black}) = -\left(\frac{1}{3} \times \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \times \log_2\left(\frac{2}{3}\right)\right) \approx 0.918$$

$$H(D_{color}) = \frac{2}{7} \times 0 + \frac{2}{7} \times 1 + \frac{3}{7} \times 0.918 \approx 0.679$$

$$IG(color) = H(D) - H(D_{color}) = 0.306$$

**Fur:**

$$H(D_{fur=ragged}) = -\left(\frac{1}{3} \times \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \times \log_2\left(\frac{2}{3}\right)\right) \approx 0.918$$

$$H(D_{fur=smooth}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D_{fur=curly}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D_{fur}) = \frac{2}{7} \times 0 + \frac{2}{7} \times 0 + \frac{3}{7} \times 0.918 \approx 0.393$$

$$IG(fur) = H(D) - H(D_{fur}) = 0.525$$

**Size:**

$$H(D_{size=small}) = -\left(\frac{3}{4} \times \log_2\left(\frac{3}{4}\right) + \frac{1}{4} \times \log_2\left(\frac{1}{4}\right)\right) \approx 0.811$$

$$H(D_{size=big}) = -\left(\frac{1}{3} \times \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \times \log_2\left(\frac{2}{3}\right)\right) \approx 0.918$$

$$H(D_{size}) = \frac{4}{7} \times 0.811 + \frac{3}{7} \times 0.918 = 0.857$$

$$IG(size) = H(D) - H(D_{size}) = 0.128$$

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**Step 2:**

$$H(D') = -\left(\frac{1}{3} \times \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \times \log_2\left(\frac{2}{3}\right)\right) = 0.918$$

**Color:**

$$H(D'_{color=brown}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D_{color=black}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D'_{color}) = 0$$

$$IG(color) = H(D') - H(D_{color}) = 0.918$$

**Size:**

$$H(D'_{size=small}) = -(1 \times \log_2(1) + 0 \times \log_2(0)) = 0$$

$$H(D'_{size=big}) = -\left(\frac{1}{2} \times \log_2\left(\frac{1}{2}\right) + \frac{1}{2} \times \log_2\left(\frac{1}{2}\right)\right) = 1$$

$$H(D'_{size}) = \frac{2}{3} \times 1 + \frac{1}{3} \times 0 \approx 0.667$$

$$IG(size) = H(D') - H(D'_{size}) = 0.251$$

5. (10 points) Design genetic algorithm to solve the Fair Treasure Split problem: Several people found a treasure that contains diamonds of different price. They need to split the treasure in n parts in such a way that the total difference in the price is minimal.

Formal definition: We have a set of numbers $S$. We need to split it into n subsets $S_1, S_2, \ldots, S_n$, such that

$$\sum_{1 \le i < j \le n} \left| \sum_{x \in S_i} x - \sum_{y \in S_j} y \right| \to \min$$

and $S_1 \cup S_2 \cup ... \cup S_n = S$, $S_i \cap S_j = \emptyset$.

(Hints: write the steps of genetic algorithm and think about how to utilize them in this problem.)

Answer:
Individual $x : x_i \in \{1, 2, \dots, n\}$
Fitness: The fitness function should reflect how close the current partition is to the optimal split. All reasonable fitness function is OK.
Initialize: number of individuals.
Crossover: take some bits from one vector, and some bits from another one.
Mutation: select one random bit and change its value.
Finish: The algorithm can stop when a fixed number of generations have been produced, or when the improvement in fitness is below a certain threshold over a number of generations.

6. (15 points) Hidden Markov Model.
   Given a hidden Markov model consisting of a box and a ball $\lambda = (A, B, \pi)$

$$A = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}, \quad B = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \\ 0.5 & 0.5 \end{bmatrix}, \quad \pi = (0.5, 0.3, 0.2)^T$$

Box state collection (box1, box2, box3)
Color of ball collection (red, white)
Let $T = 3$, $O =$ (red, white, white), compute best path.

Answer:

At $t = 1:$

$$\delta_1(1) = \pi_1 b_1(\text{red}) = 0.5 \times 0.7 = 0.35$$

$$\delta_1(2) = \pi_2 b_2(\text{red}) = 0.3 \times 0.4 = 0.12$$

$$\delta_1(3) = \pi_3 b_3(\text{red}) = 0.2 \times 0.5 = 0.10$$

$$\phi_1(1) = \phi_1(2) = \phi_1(3) = 0$$

At $t = 2:$

$$\delta_2(1) = \max(0.35 \cdot 0.6, 0.12 \cdot 0.2, 0.10 \cdot 0.3) \cdot 0.3 = 0.21 \cdot 0.3 = 0.063$$

$$\phi_2(1) = 1$$

$$\delta_2(2) = \max(0.35 \cdot 0.3, 0.12 \cdot 0.5, 0.10 \cdot 0.3) \cdot 0.6 = 0.105 \cdot 0.6 = 0.063$$

$$\phi_2(2) = 1$$

$$\delta_2(3) = \max(0.35 \cdot 0.1, 0.12 \cdot 0.3, 0.10 \cdot 0.4) \cdot 0.5 = 0.04 \cdot 0.5 = 0.02$$

$$\phi_2(3) = 3$$

At $t = 3$ :

$$\delta_3(1) = \max(0.063 \cdot 0.6, 0.063 \cdot 0.2, 0.02 \cdot 0.3) \cdot 0.3 = 0.01134$$

$$\phi_3(1) = 1$$

$$\delta_3(2) = \max(0.063 \cdot 0.3, 0.063 \cdot 0.5, 0.02 \cdot 0.3) \cdot 0.6 = 0.0189$$

$$\phi_3(2) = 2$$

$$\delta_3(3) = \max(0.063 \cdot 0.1, 0.063 \cdot 0.3, 0.02 \cdot 0.4) \cdot 0.5 = 0.00945$$

$$\phi_3(3) = 3$$

The best path is 1, 2, 2.

7. (15 points) Neural Networks.

(1) The following figure shows the architecture of a simple neural network with an input as $X = [X1, X2, X3]^T$. It includes two hidden layers, represented as $H = [H1, H2, H3]^T$, $V = [V1, V2]^T$ and the output is denoted as $Y$ (no bias used).

The activation function of the hidden and output layers takes the form of Relu $\sigma(x) = \max(0, x)$. The loss function is defined as $L(Y, T) = \frac{1}{2}(Y - T)^2$, where $T$ is target value of $Y$, the weight of input layer is $W_1$, and respectively the weights of hidden layer are $W_2, W_3$. We assume that

$$W_1 = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -2 & 1 \\ -1 & 0 & 2 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 1 & -2 \end{bmatrix}, \quad W_3 = \begin{bmatrix} 2 & -1 \end{bmatrix}$$

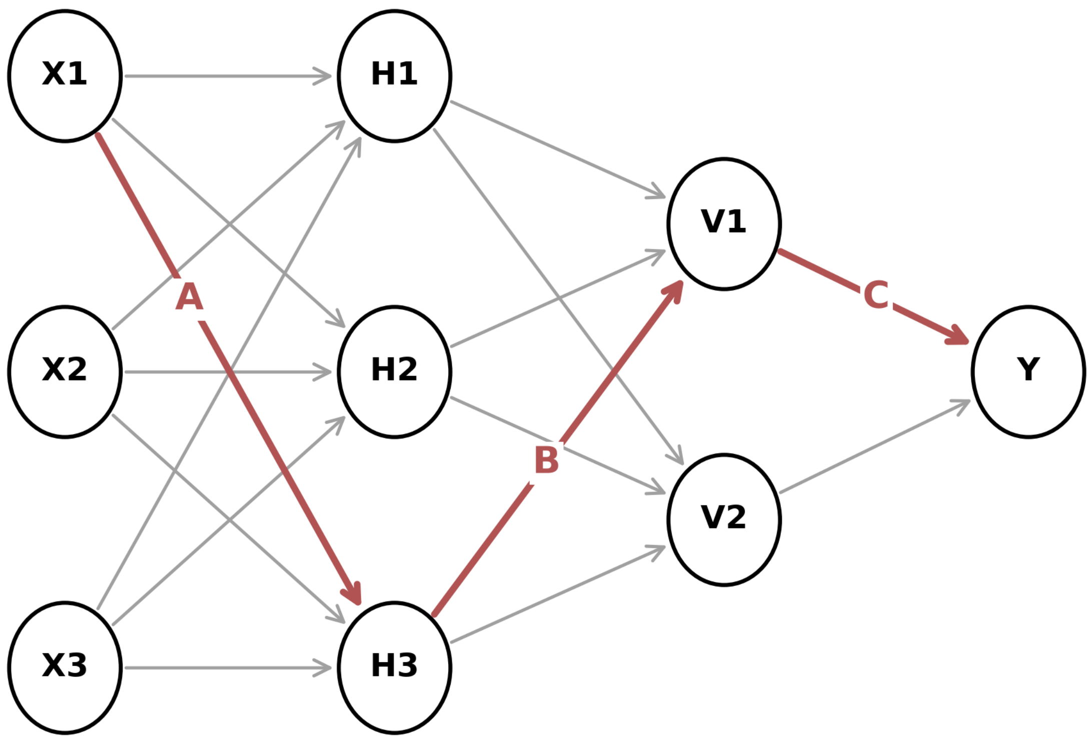


Figure 1: 第 7 题图

and $X = [1 \quad 0 \quad -2]^T, T = 1$.

(i) Please calculate the value of $Y$.

(ii) Assume that the weight edge connecting $X1$ and $H3$ is called $A$, the weight edge connecting $H3$ and $V1$ is called $B$, and the weight edge connecting $V1$ and $Y$ is called $C$. Please calculate $\frac{\partial L}{\partial A}$, $\frac{\partial L}{\partial B}$, and $\frac{\partial L}{\partial C}$.

(2) A picture is represented as

$$\begin{bmatrix} 0 & -1 & 1 & 3 \\ 1 & 0 & 2 & 1 \\ 0 & -1 & 1 & 3 \end{bmatrix}$$

the convolution kernel is represented as

$$\begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$$

the stride is 1.

(i) Please show the result after convolution operation.

(ii) Assume the pooling size is $2 \times 2$ and pooling stride is 1, please calculate the results after the max pooling and average pooling based the result above.

(iii) Please explain the reason why pooling is important.

(3) Batch normalization and layer normalization are often used in practice. Please answer why the normalization is important, and explain the differences of the batch normalization and layer normalization.

**Answer:** 注：第 1、2 小问，老师的解答是手写拍照放在了图片中，因此这里没有给出。

(3) 第一问：更好的尺度不变性，在深度神经网络中，一个神经层的输入是之前神经层的输出。给定一个神经层，它之前的神经层的参数变化会导致其输入的分布发生较大的改变。当使用随机梯度下降来训练网络时，每次参数更新都会导致该神经层的输入分布发生改变，越高的层，其输入分布会变得越明显。就像一张高墙，低楼层发生一个较小的偏移，可能会导致高楼层较大的偏移。从机器学习角度来看，如果一个神经层的输入分布发生了改变，那么其参数更新要重新学习，这种现象叫作内部协变量偏移（Internal Covariate Shift）。协变量偏移为了缓解这个问题，我们可以对每一个神经层的输入进行归一化操作，使其分布保持稳定。把每个神经层的输入分布归一化为准正态分布，可以使得每个神经层对其输入具有更好的尺度不变性。不论低层的参数如何变化，高层的输入保持相对稳定。另外，尺度不变性可以使得我们更加高效地进行参数初始化以及超参数选择。
第二问：更平滑的优化地形 (缓解梯度消失)：逐层归一化一方面可以使得大部分神经层的输入处于不饱和区域，从而让梯度变大，避免梯度消失问题；另一方面还可以使得神经网络的优化地形 (Optimization Landscape) 更加平滑，以及使梯度变得更加稳定，从而允许我们使用更大的学习率，并提高收敛速度。

批归一化是对一个批次内的数据的激活值进行归一化。层归一化是针对于单独的一条数据的激活值进行归一化。以简单的平均归一化为例子。
一个批次内的数据为 (1,2,2) (1,8,6)，批归一化后的数据为 (0.5,0.2,0.25) (0.5,0.8,0.75) 层归一化后的数据为 (0.2,0.4,0.4) (1/15,8/15,6/15)

8. (15 points) Kobe decides to train a single sigmoid unit using the following error function:

$$E(w) = \frac{1}{2}\sum_j (y(x^j, w) - y^j)^2 + \frac{1}{2}\beta \sum_j w_j^2$$

where $y(x^j, w) = s(x^j, w)$ with $s(x) = \frac{1}{1+e^{-x}}$, $s'(x) = s(x)(1 - s(x))$ being our usual sigmoid function.

(1) Write an expression for $\frac{\partial E}{\partial w_j}$. Your answer should not involve derivatives.

(2) According to (1), What update should be made to weight $w_j$ given a single training example $< x, y^* >$? Your answer should not involve derivatives.

(3) Here are two graphs of the output of the sigmoid unit as a function of a single feature $x$. The unit has a weight for $x$ and an offset. The two graphs are made using different values of the magnitude of the weight vector $\|w\|^2 = \sum_j w_j^2$.

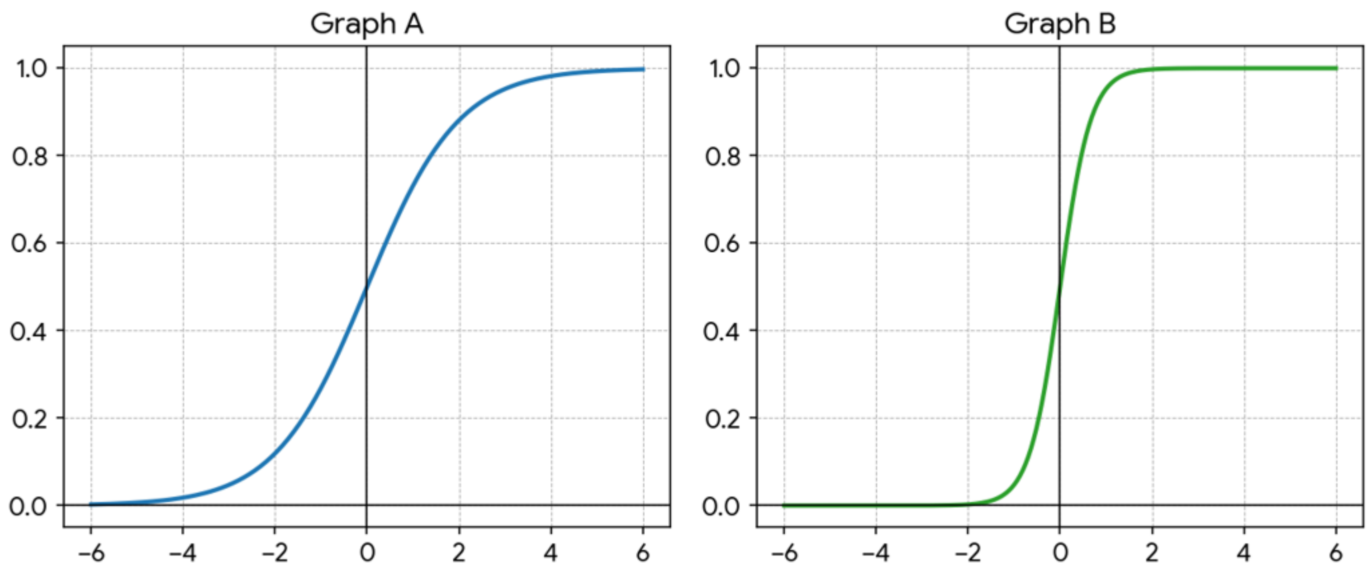


Figure 2: 第 8 题图

Which of the graphs is produced by the larger $\|w\|^2$? Explain.

(4) Why might penalizing large $\|w\|^2$, as we could do above by choosing a positive $\beta$, be desirable? How might Grady select a good value for $\beta$ for a particular classification problem?

**Answer:**

(1)

$$\frac{\partial E}{\partial w_j} = \sum (y - y^j) y (1 - y) x_j + \beta w_j$$

(2)

$$w_j' = w_j - \eta \left( (y - y^*) y (1 - y) x_j + \beta w_j \right)$$

(3) Graph B because for a given input x, the larger the magnitude of the weight vector, the greater the change in the output of the sigmoid relative to that x.
*(Note: The original problem includes two graphs (Graph A and Graph B); the answer indicates Graph B corresponds to the larger $\|w\|^2$.)*

(4) Penalizing large weight vectors would be desirable so that we avoid saturation. If the magnitude of the weight vector is too large, gradient descent will not work because the derivative of the sigmoid for large values is nearly 0. Also, large weights can cause overfitting, since they let you draw boundaries closer to the points. As we saw

with SVMs, being able to get closer to the points actually increases the size of the hypothesis class.
Grady could use cross-validation for several values of $\beta$ and use the value which yields the best cross-validation accuracy.